

2024

COMPUTER SCIENCE — HONOURS — PRACTICAL

Paper : DSCC-3P

(Data Structure)

Full Marks : 25

SET - I

Marks Distribution :

Experiment : 15

Source Code : 08

Algorithm : 04

Output : 03

Viva voce : 05

Laboratory Notebook : 05

Answer *any one* question.

1. Write a menu driven program in C to perform the following functions on an array given as input by the user :
 - (a) Print the 2nd largest number without sorting.
 - (b) Display the array in reverse order.
 - (c) Count the number of odd-elements present in even-valued array index.

2. Write a program in C to perform merging of two singly linked list. The two input lists need to be created after taking the necessary details from the user.

3. Write a program in C to implement circular queue using array, and handle all possible exceptions / error cases.

(2)

B(3rd Sm.)-Computer Sc.-H/Pr./DSCC-3P/Set-I-V/CCF

4. Write a program in C to input a list of integers, and perform the following types of sorting on them, as per user's choice —
 - (a) Quick Sort (recursive)
 - (b) Insertion Sort.

 5. Write a program in C to create a Binary Search Tree (BST) and incorporate the following operations :
 - (a) Searching for a specific item
 - (b) Performing inorder traversal (recursive)
 - (c) Delete an element.

 6. Write a program in C to implement the Push, Pop and Display operations of a stack. After this reverse the order of the first stack, using an additional stack.

 7. Write a program in C to create a doubly linked list with 'n' nodes (n → to be taken as the input from the user), and perform the following operations :
 - (a) Insert a new node at a specific position, say "pos". The validity of "pos" must be checked and handled accordingly.
 - (b) Display the list.
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